AI-Powered UTI Antibiotic System

ANALYSIS REPORT • ACADEMIC RESEARCH DOCUMENT

Patient Demographics

Age / Gender	55.0 / Male
Department	Nephrology
Diagnosis	Pyelonephritis
UTI Classification	Complicated UTI
Sample Type	Urine

Clinical Presentation

Chief Complaints: Fever;Flank pain
Comorbidities: CKD
Surgical History: None
Social History: Non smoker

Laboratory Results

Test Name	Value	Unit	Normal Range
Cbp Lymphocytes	27.0	%	20 - 40
Wbc	17400.0	cells/µL	4000 - 11000
Polymorphs	81.0	%	40 - 75
Crp	43.0	mg/L	< 10
Rft Serum Creatinine	2.2	mg/dL	0.6 - 1.3
Serum Uric Acid	6.2	mg/dL	3.5 - 7.2
Blood Urea	47.0	mg/dL	7 - 20
Cue Pus Cells	23.0	/HPF	0 - 5
Epithelial Cells	4.0	/HPF	0 - 5
Proteins	1+		Negative
Rbc	4.0	/HPF	0 - 3

AI Pathogen Prediction

Predicted Organism	**Providencia rettgeri**
Bacteria Type	Gram-negative bacillus (Associated with catheter-associated UTIs)

Antibiotic Susceptibility Profile

Predicted Sensitive	**Aztreonam**
Predicted Resistant	

Recommended Therapy

Aztreonam

Dosage: 1 g IV every 12 hours (adjusted for renal impairment; if CrCl <30 mL/min)

Precautions: Renal dose adjustment required due to CKD (serum creatinine 2.2 mg/dL). Monitor renal function and serum drug levels if prolonged therapy. Use with caution in patients with a history of beta-lactam allergy (cross-reactivity is rare but possible). Watch for emergence of super-infection (e.g., C. difficile).

Rationale: Aztreonam is the only antibiotic predicted to be sensitive against Providencia rettgeri in this isolate. It provides reliable Gram-negative coverage, including for Enterobacteriaceae, and is safe for use in patients with renal impairment when dosed appropriately.

Antibiotic Drug History

Aztreonam

Background: A unique monocyclic Î²-lactam (monobactam) approved in 1986. Its structure is distinctive because the Î²-lactam ring is not fused to another ring. Crucially, it does not cross-react with most other penicillin or cephalosporin allergies, making it a life-saving alternative for allergic patients.
Usage: Used almost exclusively for aerobic Gram-negative infections, including Pseudomonas aeruginosa, in patients with severe penicillin allergies. It is often used in combination with other drugs for intra-abdominal or pelvic infections and in inhaled form for cystic fibrosis patients.
Mechanism: Specifically targets and binds with high affinity to Penicillin-Binding Protein 3 (PBP-3) of Gram-negative aerobic bacteria. This results in the formation of long, filamentous bacterial cells that eventually lyse. It has virtually no affinity for the PBPs of Gram-positive or anaerobic bacteria.
Historical Success: Served as a critical 'niche' antibiotic that provided a safety net for patients with anaphylactic penicillin allergies who required treatment for life-threatening Gram-negative sepsis or hospital-acquired pneumonia.
Side Effects: Exceptionally low toxicity profile. Side effects are usually limited to local injection site reactions, rash, or transient elevations in liver transaminases. It is notably safe regarding the lack of cross-reactivity with most Î²-lactam allergies (except for a specific cross-reactivity with ceftazidime).
Resistance Notes: Highly susceptible to hydrolysis by Extended-Spectrum Î²-Lactamases (ESBLs) and carbapenemases. However, it is uniquely stable against Metallo-Î²-Lactamases (MBLs), leading to its experimental use in combination with avibactam to treat NDM-producing 'superbugs'.

Clinical Summary

Infection Summary

• **Type:** Complicated upper urinary tract infection (pyelonephritis) in a 55-year-old male with CKD and urinary obstruction.
• **Predicted pathogen:** *Providencia rettgeri* (Gram-negative bacillus, often catheter-associated).

Antibiotic Susceptibility

• No resistant agents were predicted.
• **Sensitive:** Aztreonam is the only drug predicted to retain activity against this isolate.

Recommended Therapy

• **Aztreonam 1 g IV every 12 h -** dose to be reduced for renal impairment (adjust if CrCl < 30 mL/min).
• **Rationale:** Aztreonam provides reliable coverage of Gram-negative Enterobacteriaceae, including *Providencia* spp., and is the sole agent predicted to be effective. It is safe in renal disease when appropriately dosed and avoids cross-reactivity with most β-lactam allergies.

Precautions & Monitoring

• Adjust dose for the patient's elevated serum creatinine (2.2 mg/dL).
• Monitor renal function and, if therapy extends beyond several days, consider serum drug levels.
• Observe for allergic reactions (rare β-lactam cross-reactivity) and for super-infection such as *Clostridioides difficile* colitis.

Actionable Plan

Initiate aztreonam with renal-adjusted dosing, reassess renal parameters and clinical response within 48-72 h, and modify therapy only if new susceptibility data or adverse events emerge.